



Multimodal fall detection framework: Leveraging bed-integrated load cells and wrist accelerometers

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ABSTRACT

Falls are the leading cause of injury and mortality among the elderly, necessitating reliable detection systems for timely intervention. Existing fall detection methods often rely on wearable devices or vision-based systems, each with unique strengths and practical considerations related to user comfort, battery life, and continuous monitoring capability. This paper presents a hybrid fall detection system that integrates contactless load cells on a bed frame and a wrist-mounted accelerometer. This proposed detection system integrates the advantages of existing fall detection methods. Data were collected from 40 healthy adults performing simulated Fall and Non-fall scenarios in six postures. A one-dimensional convolutional neural network was developed and trained using three configurations: load cell data only, accelerometer data only, and a combination of both. The accuracy, precision, and sensitivity of the system were evaluated. Signal analysis, including frequency-domain assessments, identified distinguishing features in load cell signals. The hybrid model achieved 96.26% accuracy, 95.21% precision, and 97.27% sensitivity, outperforming single-sensor models by 21.82%. Frequency analysis revealed that low-frequency bands (0–2 Hz) in load cell signals were critical for differentiating Falls from Non-falls. High-performing participants exhibited stronger spectral power in this range, highlighting their role in fall detection. Although fully contactless systems are a long-term goal, hybrid systems offer practical intermediate solutions. Our findings highlight the potential of hybrid fall detection systems to improve accuracy and identify critical low-frequency biomarkers, facilitating the advancement toward fully contactless solutions.

1. Introduction

Falls are a leading health concern, particularly among the elderly, and declining physical ability increases the risk of serious injury or death [1,2]. According to the World Health Organization, approximately 172 million people worldwide suffer serious injuries from falls annually, with approximately 684,000 fall-related deaths reported annually [3]. Elderly individuals are subjected to risk of falls due to age-related muscle weakness and impaired mobility [4,5]. Falls can lead to prolonged recovery periods and reduced quality of life, along with psychological consequences like fear of falling, which can limit daily activities and contribute to social isolation and depression [6,7]. Given these risks, early and reliable fall detection systems are essential in elderly care to enable timely medical intervention and minimize negative outcomes.

Fall detection systems can be categorized using various criteria, such as the type of sensor, data processing methods, and device characteristics [8,9]. In this study, we adopt a hierarchical approach based on the type and characteristics of data acquisition devices, dividing systems into wearable and non-wearable categories. Non-wearable devices are further subdivided into vision-based and non-vision sensor systems. Further, non-vision sensor-based systems can be categorized into pressure sensors, infrastructure-based accelerometers, and radio-frequency identification (RFID) sensors.

Wearable devices, such as smartwatches and accelerometer-based sensors, are widely used to monitor movements and detect falls with high accuracy [10–12]. These systems benefit from low cost and ease of integration into daily life. However, they require regular charging and continuous wear, which may reduce user adherence, particularly during sleep.

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By contrast, non-wearable fall detection systems can be divided into vision- and non-vision-based systems installed in the environment. Vision-based systems leverage advancements in image processing to provide high-precision fall detection, but pose privacy concerns and processing challenges because of large video data volumes [13–15].

Non-vision-based systems typically include RFID, floor-mounted accelerometer, and pressure-sensor-based approaches. RFID-based methods use tags worn by users and detected by readers to monitor movements. However, this method is not fully contactless, as users must wear tags, and may be susceptible to interference from other electronic devices [16,17].

Pressure sensor systems have shown promise among non-vision methods due to advantages like contactless monitoring, privacy protection, and suitability for continuous monitoring in homes and hospitals. These sensors embed pressure-sensitive elements in surfaces including floors, carpets, or mattresses to detect pressure distribution changes from body movements [18–20]. Recent advances include smart carpets using air pressure sensors to analyze variations [18], fiber-optic sensor-embedded carpets for posture recognition and fall classification [19], piezoelectric sensor arrays in floors for capturing subtle dynamic pressure signals [20], mats with force-sensitive resistors to monitor pressure distributions [21], and mattress-based whole-torso pressure detection systems to identify risky postural shifts [22].

Despite these promising results, existing pressure sensor-based systems face several limitations. First, array- or mattress-type systems often require extensive sensor deployment to cover large areas, increasing cost and complexity, and may struggle to detect falls if the user moves out of the monitored area. Many pressure sensor systems rely on threshold-based algorithms, which may not be sufficiently flexible to accommodate variations in fall patterns or different types of users. Moreover, pressure sensors embedded in mattresses or bed frames raise concerns about durability and maintenance, especially in real-world applications where sensors may wear out from constant use or require frequent recalibration. Second, despite the potential advantages of load cells, such as durability and minimal maintenance requirements, this type of pressure sensor remains largely unexplored for fall detection. To date, preprocessing, feature extraction, and analysis of load cell signals have been rarely explored for fall detection. Critical aspects like identifying effective frequency bands and specific signal processing techniques for load cell data remain unexplored, hindering the maximization of detection accuracy in load cell-based systems. The high prevalence of falls in bed-related environments is a critical issue that has not been sufficiently addressed. According to Ibrahim et al. [8], 60%–70% of hospital falls occur near or around the bed, and 65.9% of home falls occur in the bedroom, with 80.1% specifically around the bed. Floor-based systems have been extensively studied, whereas few studies have focused on bedside fall detection despite falls in these areas being frequent and posing significant health risks. Therefore, research on bedside fall detection should be prioritized to better reflect real-world scenarios. Moreover, the feasibility of integrating pressure sensors with other modalities, such as wearable devices or vision-based systems, has remained underexplored. Multimodal systems combining data from multiple sensors could improve fall detection accuracy and reduce false positives [23]. However, studies on combining pressure sensors with other technologies to enhance fall detection performance are scarcely reported.

The primary objective of this study was to validate the efficacy of load cell-based fall detection for bedside monitoring, as an intermediate step toward achieving fully contactless systems. The proposed system integrates bed-mounted load cells with wrist-worn accelerometers, leveraging their complementary strengths, balanced detection accuracy, user convenience, and practicality. Specifically, this study addresses the critical gaps in existing approaches via the following unique contributions:

1. **Hybrid sensing framework:** Proposing and validating an innovative multimodal approach that systematically combines bed-integrated load cell signals with wrist-mounted accelerometer data, effectively harnessing their complementary strengths for enhanced fall detection accuracy.
2. **Validation of load cell efficacy for bedside monitoring:** Demonstrating the utility and practicality of load cells as robust, unobtrusive, and cost-effective sensors for bedside fall detection, especially suited for large-scale hospital and home-care environments.
3. **Identification of low-frequency spectral biomarkers:** Revealing critical low-frequency (0–2 Hz) spectral signatures in load cell data that differentiate fall events from normal bed movements, providing new biomarkers for potential early fall prediction and detection models.
4. **Optimization and interpretability of convolutional neural network (CNN)-based fall classification:** Developing a one-dimensional CNN (1D CNN) architecture optimized for multimodal sensor integration, with detailed evaluation and interpretation of frequency-dependent performance, thus enhancing model transparency and clinical applicability.
5. **Insights into personalized fall detection strategies:** Exploring interindividual variations in sensor-derived features and performance, laying the groundwork for future adaptive, personalized fall detection and monitoring approaches tailored to individual behavioral patterns and sensor characteristics.

2. Related works

Fall detection systems have been widely studied and typically classified based on the sensing technologies employed, including wearable sensors, vision-based systems, and non-vision sensor-based methods. Each category provides distinct advantages and faces unique challenges during practical deployment.

Wearable-based systems utilize inertial sensors like accelerometers and gyroscopes to monitor user movement, typically embedded in worn devices. Chen et al. used a single accelerometer-based wearable to classify eight daily activities, including falls, achieving 93.8% accuracy using a CNN [24]. Wang et al. integrated a tri-axial accelerometer and barometric sensor in a wrist-worn device to detect falls, achieving sensitivity of 91% and a false positive rate of 0.1149 in low-power environments [25]. Purushothaman et al. collected data from wearable sensors but did not specify performance metrics [26]. Quadros et al. developed a wrist-worn device with an inertial measurement unit (IMU) to classify six fall scenarios and six daily activities using k-nearest neighbors, achieving 99% accuracy and high sensitivity and specificity [27]. Hashim et al. proposed a fall detection system for Parkinson's disease patients using wearable accelerometers, electromyography sensors, and a ZigBee wireless network, achieving 100% accuracy under optimal threshold conditions [9]. These systems offer high detection performance, portability, and ease of integration; however, they are limited by user adherence, particularly during sleep, and require regular charging.

Vision-based fall detection systems use optical, depth cameras, and radar sensors to capture posture and movement without wearable devices. Zhao et al. utilized depth cameras to detect falls based on user's center of mass, achieving high accuracy with a random forest (RF)-based upper body detection and binary fall classifier [28]. Su et al. applied a CNN-based model to depth video data, including in-bed fall scenarios, reporting 95% accuracy and a Cohen's kappa coefficient of 0.96 [29]. Vision-based systems provide contactless monitoring without user adherence, but raise privacy concerns and require significant computational resources due to video data volume.

Non-wearable, non-vision sensor-based systems include ambient sensing using RFID tags and embedded accelerometers or vibration sensors. It is referred to as ambient sensor [9], classified distinctly from

directly installed devices based on configuration and usage [8]. Wickramasinghe et al. developed a hybrid RFID and accelerometer-based device worn on the sternum and used support vector machines (SVM) to achieve over 90% accuracy, sensitivity, and specificity [30]. Zhu et al. proposed an RFID tag-based system worn around the neck, achieving 98% true positive rate for classifying seven activities, including falls, using an SVM [16]. Ibrahim et al. installed accelerometer-based sensor nodes around a mattress perimeter to detect falls through vibration patterns, reporting 90.3% accuracy with a rule-based classifier [31]. Clemente et al. utilized seismometers on the floor to identify falls via footstep-induced vibrations, achieving 73.0% accuracy using an SVM [32]. These ambient systems provide contactless monitoring but face limitations in sensor deployment complexity, environmental noise sensitivity, and lower accuracy versus other methods.

Pressure sensor-based systems present another promising non-vision ambient modality, employing sensors embedded within surfaces such as carpets, mats, and mattresses to monitor movements by detecting subtle pressure changes. They have the potential to offer a fully contactless and privacy-preserving solution. Several studies have employed pressure sensors to develop fall detection systems, typically categorized into detecting falls on the floor and bedside falls. Floor-based systems often employ sensor-embedded carpets or mats to detect falls based on force distribution. Zhang et al. used passive infrared sensors with piezoelectric microphones to detect falls based on movement patterns, achieving 88.9% accuracy, though requiring extensive sensor coverage [33]. Chaccour et al. developed a smart carpet with air pressure sensors using a rule-based algorithm identifying falls when the threshold was exceeded in three of four sensors, reporting 88.8% sensitivity and 94.9% specificity [18]. Feng et al. designed a fiber-optic sensor-embedded carpet that classified postures into lying, sitting, or standing, achieving 97% accuracy in fall detection based on lying posture duration [19]. Minvielle et al. embedded piezoelectric sensors across room floors, extracted 51 features from sensor data using first derivatives and Fourier transforms, and achieved 98.4% accuracy with a RF classifier [20]. Yeo et al. integrated force-sensitive resistors into a mat, using a shape-detection algorithm for posture classification and a color histogram-based algorithm for fall detection, achieving 80% and 90% accuracy, respectively [21]. Kowshik et al. proposed combining wearable accelerometers with piezoelectric sensors in bathroom floors, though this system lacked experimental validation [34]. While effective in larger spaces, these systems often require numerous sensors for full room coverage, increasing costs and limiting utility for detecting falls in confined spaces or in-bed scenarios.

In contrast to floor-based systems, bedside fall detection systems target a smaller area such as the bed and its immediate surroundings. These systems address a critical issue: falls often occur when individuals attempt to get in or out of bed. Youngkong et al. developed a pressure sensor system embedded in a mattress and surrounding floor area using 144 pressure-sensitive units on the mattress and 72 units on the floor [35]. The system achieved 100% accuracy in detecting falls by monitoring posture changes and classifying fall scenarios using a RF classifier. Similarly, Wafiq et al. employed force-sensitive resistors within a mattress to monitor the user's position and detect dangerous movements or prolonged absence from the bed [36]. Using a rule-based algorithm, the system detected bed exits with 92% accuracy. Zhao et al. introduced a whole-torso distributed pressure detection mattress that calculated the user's center of mass and flagged risky situations when the user's body extended beyond a predefined safety zone [22]. The system detected zero-missed falls in a trial involving 30 participants.

In summary, while wearable and vision-based methods dominate the field because of their detection performance and flexibility, pressure sensor-based systems offer a promising, underexplored avenue for achieving contactless, privacy-aware, and low-maintenance fall detection. Their potential for hybrid integration with wearable sensors may further enhance detection accuracy and robustness.

3. Materials and methods

Fig. 1 provides a visual representation of the overall experimental workflow, outlining the data collection setup, and an example of the acquired signal. To acquire the data, load cells were installed on the four legs of the bed, and an accelerometer was placed on the participant's left wrist, as shown in Fig. 1(a). Fig. 1(b) shows sample signals from the load cell and accelerometer collected during these actions. The left two graphs show signals from the load cell data from the four channels, and the right graphs show accelerometer signals from the X, Y, and Z channels. The shaded areas indicate the movement phases. Additionally, the top graphs show signals from the simulated "Fall", and the bottom graphs representing "Non-fall" conditions. The experimental procedure, depicted in Fig. 1(c) and (d), involved participants performing two key actions: (c) safely exiting the bed ("Non-fall") and (d) simulating a fall ("Fall"). Fig. 1(e) illustrates the flow of the fall detection procedures, including data acquisition, preprocessing, model learning, and subsequent fall detection.

3.1. Devices

We employed a multimodal approach for fall detection utilizing both contactless load cells and wearable accelerometers. The load cells were placed under the four legs of a standard hospital bed, and the participants wore a wrist-mounted accelerometer to capture movement data.

We used FX1901-0001-0100-L (TE Connectivity, USA) load cells capable of detecting forces up to 100 lb. The sensors output an analog voltage signal between 0 and 5 V, which was converted to 0–3.3 V using the JY-S65 amplifier module (Shenzhen Entway Electronic Components Co., China). These amplified signals were fed into an analog-to-digital converter (Nucleo-F103RB microcontroller board, STMicroelectronics, Switzerland), which featured an STM32F103RB microcontroller unit. The analog-to-digital converter was operated at 12-bit resolution with a sampling rate of 100 Hz.

The wearable device used was a BioNomadix Wireless Accelerometer Transmitter (BIOPAC Systems, Inc., USA), which recorded three-axis acceleration data at a sampling rate of 1000 Hz. The data were transmitted to an MP160 acquisition system (BIOPAC Systems, Inc., USA), where they were processed and stored using the Acknowledge software platform (BIOPAC Systems, Inc., USA).

Synchronization between the load cells and accelerometer was achieved using a hardware-triggered synchronization system. Specifically, the synchronization system used three wired connections: a trigger line for synchronizing data collection start/end, and two labeling lines for synchronizing event labels. These connections linked three general-purpose input–output pins from the Nucleo-F103RB microcontroller board to three analog input channels (Channels 1–3) of the MP160 acquisition system. The trigger line used Channel 1, initiating data collection upon detecting 3 V or higher. When collection started, a 3 V pulse through this pin triggered data collection across both systems. Channels 2 and 3 were used to digitally label the three experimental conditions based on voltage states. Further details on labeling schemes are provided in Section 3.2.2.

To ensure data accuracy and reliability, both the load cell and accelerometer underwent calibration and verification procedures. For the load cell, a five-point static calibration was performed using a certified 10 kg standard weight (Ivanko Barbell Company, USA), covering the expected force range (0–40 kg). The calibration was conducted in the same environment as the experiments, under controlled indoor conditions at room temperature (25 °C), to prevent temperature differences between calibration and experiments. Each reference weight was applied repeatedly, and voltage outputs were statistically validated. For the accelerometer, a gravity-based orientation test confirmed zero-offset and scale fidelity along all three axes. The ± 1 G (gravitational acceleration, $1 \text{ G} \approx 9.807 \text{ m/s}^2$) test confirmed the expected $\pm 0.625 \text{ V}$

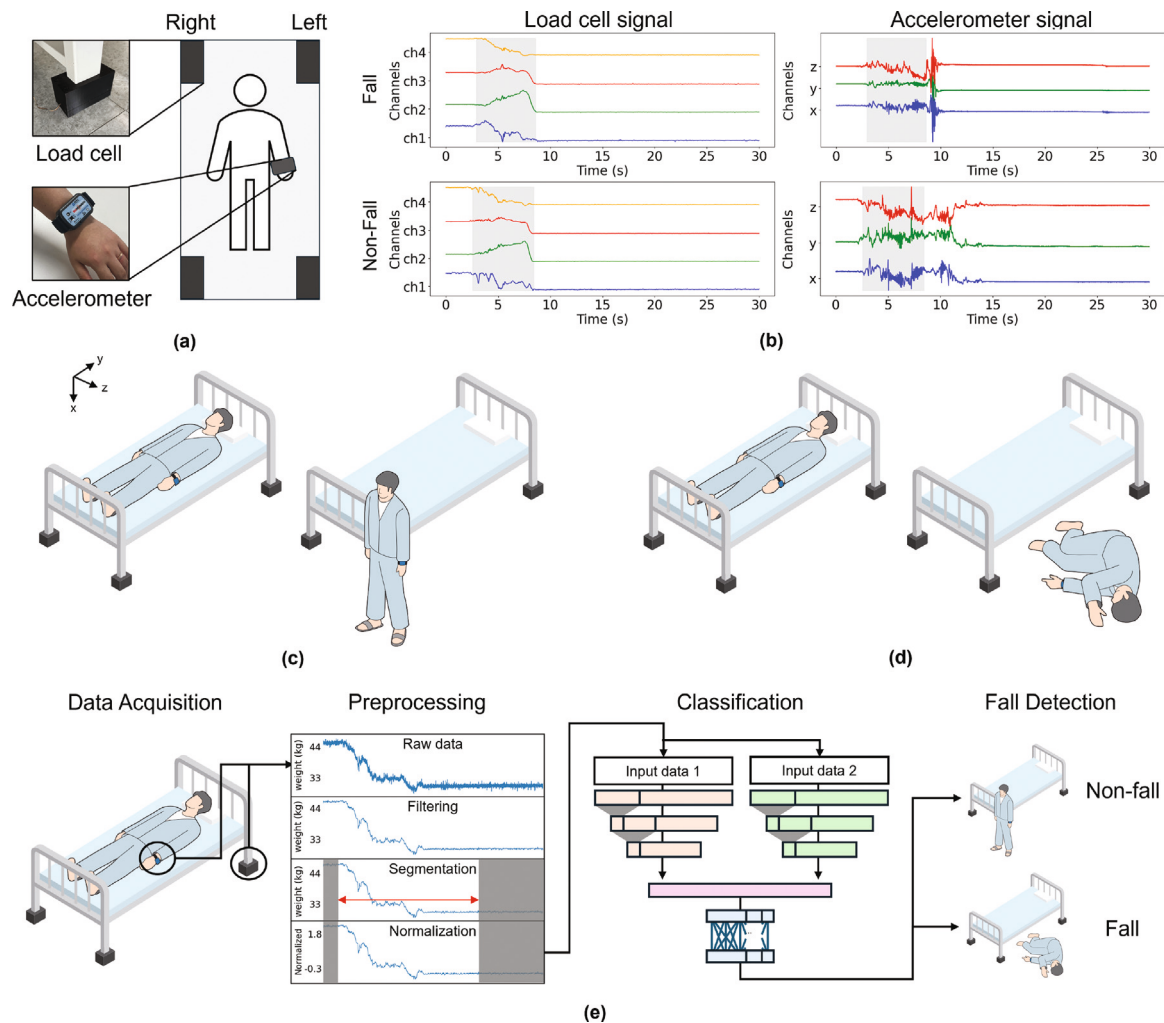


Fig. 1. Overall illustration of the experimental setup and data collection process. (a) Placement of load cells on the bed legs and accelerometer worn on the participant's wrist; (b) Example of collected signals; (c) Visualization of the condition: Non-fall; (d) Visualization of the condition: Fall; (e) Flowchart of the fall detection algorithm.

outputs in the ± 16 G (± 10 V) range for the X, Y, and Z axes. These validation procedures were repeated to ensure repeatability and sensor responsiveness. Detailed calibration setups, procedures, and results for both sensors are provided in the supplementary material.

3.2. Data collection

3.2.1. Recruitment

This study was conducted according to the principles outlined in the Declaration of Helsinki and approved by the Institutional Review Board of Hallym University (HIRB-2024-012). Forty healthy adults (20 males and 20 females) aged 19 years and above, with no cardiovascular impairments or movement limitations, participated in the experiment. The mean age of the participants was 23 ± 1.78 years, with an average height of 168.4 ± 10.3 cm and an average weight of 63.9 ± 13.6 kg. All participants provided written informed consent before participating in the experiments.

The participants were provided with detailed information regarding the study procedure, including any potential risks. A detailed explanation of the experimental procedure was provided before the trial began. Participants were instructed to wear comfortable clothing during the experiment. Upon arrival, the participants read the study information sheet, signed the consent form, and completed a questionnaire that included demographic details such as age, height, and weight. The participants wore an accelerometer on their left wrist.

3.2.2. Experimental details

The experiment was designed to collect data for both “Fall” and “Non-fall” events during transitions in and out of bed. The participants were instructed to simulate both a controlled fall and normal bed exit under supervised conditions. In the fall scenario, the participants moved from a lying position to sitting on the edge of the bed before simulating sudden loss of balance and falling. A Non-fall event involved getting out of bed.

Each participant performed these actions in six different postures: supine, supine with arms spread, left lateral, right lateral, left semi-lateral, and right semi-lateral. These varied postures in the laydown phase were selected to ensure the robustness of the fall detection system by simulating real-life scenarios in which individuals may fall from different positions.

Each trial lasted 155-s, consisting of a 5-s preparation phase, a 120-s task phase (including lying down and performing either a Fall or Non-fall action), followed by a 30-s rest phase. At the beginning of each trial, the participants were instructed to stand approximately 1 m from the bed. Once data acquisition was triggered by the experimenter, the participants were given a 5-s countdown before being instructed to lie down in the specified posture. Afterward, the participants lay down on the bed for 30-s (lay down). After the laydown action, the participants were instructed to perform a non-fall action. For the Non-fall actions, they stepped out of the bed normally and stood in the starting position. This process was repeated for the fall action over

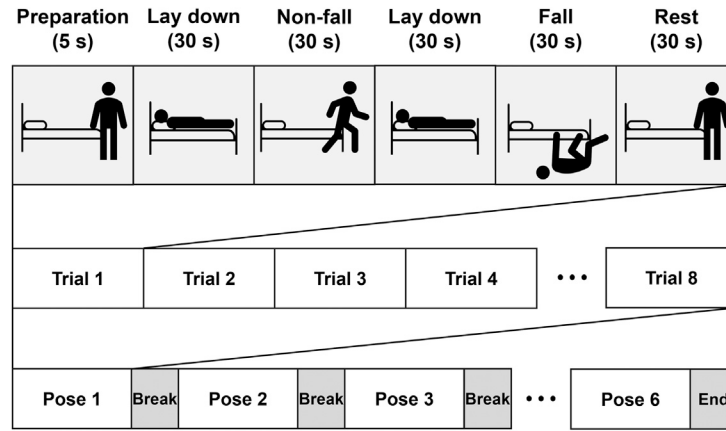


Fig. 2. Proposed experimental paradigm.

the next 60-s. During the fall actions, the participants were requested to remain in the fallen position for 30-s. After the task phase, the participants rested comfortably on the bed for 30-s to prepare for the next trial (rest phase). A safety mattress was placed around the bed to ensure the safety of participants during the fall simulations.

To label the experimental conditions within the collected data, two analog input channels (2 and 3) of the MP160 system were used as digital event markers to label experimental conditions. These signals were generated by the Nucleo-F103RB microcontroller through general-purpose input–output pins. The labeling scheme was: when Channels 2 and 3 were 0 V (off), it indicated “Lay down” action; when Channel 2 was 3 V (on) and Channel 3 was 0 V (off), it indicated “Non-fall” action; when Channel 2 was 0 V (off) and Channel 3 was 3 V (on), it indicated “Fall” action. This digital labeling enabled precise segmentation and identification of experimental events within the synchronized data streams. The experimental protocol was repeated eight times per posture across all six postures, resulting in 96 actions (48 Falls and 48 Non-falls) per participant. To prevent fatigue, a 5-min break was provided between trials. The detailed experimental procedure is shown in Fig. 2.

3.3. Preprocessing and data splitting

The data collected from the load cells and accelerometers were preprocessed to remove high-frequency noise and normalize the signals. Because of the limited research on fall detection using load cells, the effective frequency range for capturing movement signals from load cells has not been well established. Therefore, various frequency band combinations were investigated to identify the optimal frequency bands. A 5th-order Butterworth filter was applied with different configurations, such as low-pass, high-pass, and bandpass filters, covering frequencies of 0–50 Hz. This filtering process was applied to both load cell and accelerometer data.

Following the filtering process, the signals were segmented into 30-s windows based on hardware triggers that marked the onset of each trial. The load cell data sampled at 100 Hz were divided into 3000 sample windows, whereas the accelerometer data sampled at 1000 Hz were segmented into 30,000 sample windows. Manual inspection was performed to remove artifacts, including contaminated segments and outliers. Following this step, the data were normalized to ensure amplitude consistency across all trials and participants. Of the initial 3840 segments collected from 40 participants (96 segments per participant), 3803 segments were retained after preprocessing.

An interparticipant scheme was employed for data splitting. The dataset was split into training and test sets using five-fold cross-validation. Each fold consisted of data from 32 participants for training and data from the remaining 8 participants for testing, ensuring that the model was evaluated on unseen subjects to validate its generalizability.

3.4. Fall detection algorithm

A 1D CNN was developed to classify Fall and Non-fall events based on collected sensor data. Three models were trained using different combinations of input data: (1) contactless load cell data (batch size $\times$ 4 sensors $\times$ 3000 time samples [30-s at 100 Hz]), (2) wearable accelerometer data (batch size $\times$ 3 axes (X, Y, Z) $\times$ 30 000 time samples [30-s at 1000 Hz]), and (3) a multimodal approach that combined load cell and accelerometer data.

The architecture of the proposed 1D CNN consisted of two main types of convolutional blocks: one with max pooling (Block 1) and one without max pooling (Block 2). Each block includes convolutional layers with rectified linear unit (ReLU) activation, batch normalization, and dropout. In Block 1, max pooling was applied to downsample the feature maps, whereas in Block 2, the same structure was used but without pooling. These blocks were alternately stacked to extract the features.

Specifically, the initial convolutional layer had 16 filters with kernel size 5, and subsequent layers featured channel sizes of 32, 64, 128, and 64 with kernel sizes of 9, 17, 33, and 17, and padding sizes of 2, 4, 8, 16, and 8. Stride was 1 for all convolutional layers, except the fifth block with stride 2. Max pooling layers followed convolutional layers in Block 1, with pooling sizes of 2, 2, and 5. After the final convolutional block, a global average pooling layer reduced the temporal dimension to a single value per channel.

After the convolutional blocks, the network includes two fully connected layers with a ReLU and sigmoid activations. The first fully connected layer utilized an ReLU activation function, followed by a dropout layer, whereas the final layer used a sigmoid activation function for binary classification. The final output layer provides a binary classification, where 0 indicates a non-fall event and 1 indicates a fall. The detailed structure is shown in Fig. 3.

The model was trained using the Adam optimizer with a learning rate of 0.0001, and binary cross-entropy was employed as the loss function. A batch size of 32 was used, with a dropout rate of 0.2 applied across all layers to prevent overfitting. The maximum number of epochs was set to 300, and early stopping was triggered if no improvement in validation performance was observed for 100 consecutive epochs. To ensure reproducibility, global random seeds were fixed throughout the experiments. Specifically, the Python hash seed (PYTHONHASHSEED), Python's `random.seed()`, NumPy's `numpy.random.seed()`, and PyTorch's `torch.cuda.manual_seed()` were consistently set to 42. Additionally, local functions involving randomness, such as the `GroupKFold` from scikit-learn, were configured with a `random_state` set to 42. Deterministic behavior was further ensured by setting `torch.backends.cudnn.deterministic = True` and `torch.backends.cudnn.benchmark = False`. The model's performance

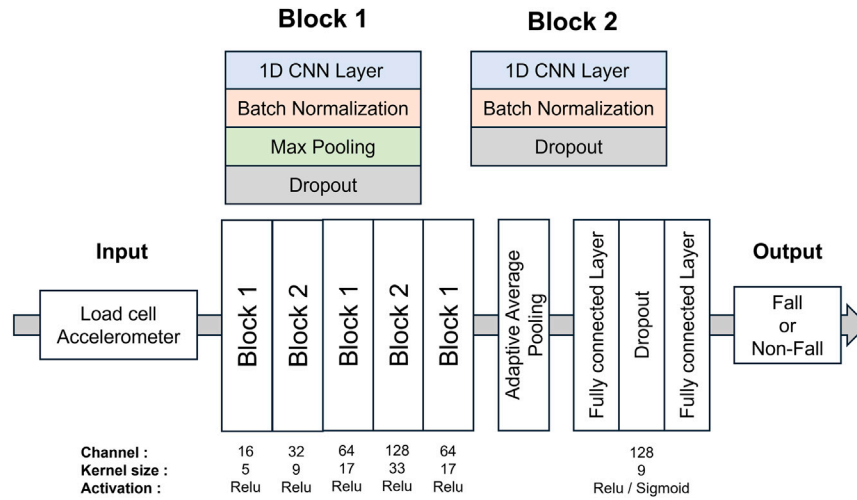


Fig. 3. Structure of the proposed 1D CNN.

was evaluated using accuracy, precision, and sensitivity metrics to assess its effectiveness for fall detection.

For the multimodal approach, which utilizes both load cell and accelerometer data, a late fusion strategy was employed. Sensor inputs were independently processed through identical convolutional blocks, compressing load cell and accelerometer signals into 64-dimensional feature vectors after global average pooling. The feature vectors from each sensor were concatenated to form a unified 128-dimensional representation. This combined vector passed through fully connected layer blocks identical to single-sensor models, generating binary classification results.

3.5. Signal analysis

This study aimed to identify fall-related components in load cell signals to identify potential biomarkers for fall risk prediction, early detection, and performance optimization. Although accelerometer-based fall detection has been studied previously, this study specifically focused on load cell signals, given their relatively unexplored applications in fall detection. Therefore, the accelerometer signals were not included in the analysis. Signal analysis was conducted in three stages: (1) comparison of Fall and Non-fall signals in both frequency and time domains; (2) group-based comparisons of high- and low-performing participants in fall detection accuracy to explore the contributing factors; (3) and correlation analysis between model performance and participant or signal characteristics.

First, a frequency-domain analysis was conducted to identify significant differences between the Fall and Non-fall signals. Load cell signals were filtered using a 0–10 Hz bandpass filter, and the power spectral density (PSD) was computed on a logarithmic scale for each signal using a fast Fourier transform. Statistical comparisons were performed between the Fall and Non-fall signals using paired t-tests for each frequency bin. To address the potential for false positives due to high dimensionality, Bonferroni correction was applied to adjust for multiple comparisons. This analysis aimed to identify the most critical frequency ranges for distinguishing between falls and non-falls. In addition to frequency-domain analysis, we performed a time-domain analysis to characterize temporal differences between Fall and Non-fall load cell signals during bed-exit transitions. Load cell signals from Fall and Non-fall events were visually and statistically compared. A paired t-test was conducted at each time point between Fall and Non-fall signals across four load cell sensors at the bed frame legs. Intervals showing significant differences ($p < 0.05$) were identified to determine timing

and characteristics of load cell signal divergence between Fall and Non-fall events. This analysis was conducted to reveal the distinct load cell signal features that could enhance fall detection.

Second, the participants were grouped into three categories based on their fall detection accuracy to investigate the factors influencing the classification performance. Thresholds were determined using the five-fold average accuracy of fall detection (74.44%) with a $\pm 5\%$ range, ensuring balanced group sizes while maintaining statistical robustness in performance comparisons. Based on these thresholds, participants with an accuracy above 79.44% were assigned to the high-performing group (Group 1), those with an accuracy between 69.44% and 79.44% were categorized into the intermediate-performing group (Group 2), and those below this range were assigned to the low-performing group (Group 3). For further analysis, Group 2 was excluded to enable a clearer differentiation of the factors between high- and low-performing participants. PSD distributions of the Fall and Non-fall signals were computed separately for each group, and paired t-tests with Bonferroni correction were used to determine significant spectral differences. Additionally, between-group comparisons were performed to assess differences in PSD characteristics (e.g., Group 1 fall vs. Group 3 Fall, Group 1 fall vs. Group 1 non-fall). This analysis aimed to identify distinguishing spectral features contributing to the high classification performance in Group 1 and explore potential predictive markers for performance screening using load cell signals.

Third, to investigate the factors influencing classification performance, a Pearson correlation analysis was performed between performance metrics (e.g., accuracy) and participant characteristics, such as body mass index (BMI), height, and PSD values across specific frequency bands. This analysis aimed to identify the correlations between participant demographics or spectral features and model accuracy, highlighting factors that could inform model optimization strategies.

4. Results

4.1. Load cell

Table 1 presents the averaged five-fold cross-validation detection performance across frequency bands applied to load cell signals. The highest accuracy ($74.44 \pm 3.21\%$) and specificity ($82.99 \pm 3.85\%$) were observed in the 0–10 Hz band. However, the highest sensitivity ($65.71 \pm 6.83\%$) was recorded in the 0–50 Hz band. Bands including the 0–10 Hz range consistently achieved higher accuracies. Excluding this range leads to a significant decrease in accuracy, demonstrating the importance of low-frequency components in load cell signals for fall detection.

Table 1

Averaged five-fold cross-validation performance for various frequency bands of load cell signals (mean $\pm$ standard deviation for test data).

Band (Hz)	Accuracy (%)	Sensitivity (%)	Specificity (%)
0–10	74.44 $\pm$ 3.21	61.61 $\pm$ 04.89	82.99 $\pm$ 3.85
0–20	74.17 $\pm$ 3.20	63.15 $\pm$ 12.46	81.91 $\pm$ 3.87
0–30	73.78 $\pm$ 2.66	63.18 $\pm$ 06.93	80.76 $\pm$ 4.96
0–40	73.52 $\pm$ 2.64	64.08 $\pm$ 05.97	79.03 $\pm$ 1.68
0–50	74.33 $\pm$ 3.54	65.71 $\pm$ 06.83	79.38 $\pm$ 2.50
10–20	54.16 $\pm$ 0.54	59.94 $\pm$ 19.40	54.56 $\pm$ 2.30
10–30	55.06 $\pm$ 0.57	50.02 $\pm$ 14.31	56.21 $\pm$ 1.72
10–40	54.42 $\pm$ 1.11	56.24 $\pm$ 16.37	54.54 $\pm$ 1.05
10–50	53.95 $\pm$ 0.51	70.92 $\pm$ 19.48	53.55 $\pm$ 1.92
20–30	53.14 $\pm$ 1.28	42.48 $\pm$ 22.56	55.23 $\pm$ 2.26
20–40	53.19 $\pm$ 1.07	55.64 $\pm$ 24.37	53.66 $\pm$ 1.61
20–50	52.42 $\pm$ 0.70	50.45 $\pm$ 15.89	52.61 $\pm$ 0.27
30–40	53.72 $\pm$ 0.89	63.36 $\pm$ 24.64	54.77 $\pm$ 4.02
30–50	52.48 $\pm$ 0.82	38.40 $\pm$ 27.97	56.38 $\pm$ 3.92
40–50	53.27 $\pm$ 1.05	59.77 $\pm$ 18.76	53.09 $\pm$ 0.96

4.2. Accelerometer

Table 2 shows the performances of models trained with accelerometer signal following the same setup as the load cell analysis. The highest accuracy (95.05 $\pm$ 1.18%), sensitivity (95.64 $\pm$ 2.46%) were achieved in the 0–50 Hz band, whereas specificity peaked at 97.06 $\pm$ 2.83% in the 10–40 Hz band. This suggests accelerometer signals benefit from retaining a broader frequency range for accurate classification, capturing both slow and rapid motion changes. Notably, accelerometer-based models remained relatively stable across different frequency ranges and generally outperformed load cell models. The accuracy reduction when excluding the 0–10 Hz band was approximately 3% compared with the 20% drop observed with the load cell signal. This result indicates that while the low-frequency range is crucial, accelerometer signals with low sensitivity to this range facilitate effective fall detection. The findings highlight the robustness of accelerometer data, with the 0–50 Hz range being optimal for maximizing detection performance.

4.3. Multimodal approach

Table 3 shows that models trained using only load cell data achieve an accuracy of 74.44 $\pm$ 3.21%, sensitivity of 61.61 $\pm$ 4.89%, and specificity of 82.99 $\pm$ 3.85%, which are lower than those of other configurations. The model using wrist-mounted accelerometer data showed higher performance, achieving an accuracy of 95.05 $\pm$ 1.18%, sensitivity of 95.64 $\pm$ 2.46%, and specificity of 94.58 $\pm$ 1.66%. The multimodal model achieved the highest performance with an accuracy of 96.26 $\pm$ 1.64%, sensitivity of 97.27 $\pm$ 1.48%, and specificity of 95.21 $\pm$ 2.81%. The best frequency bands (0–10 Hz for the load cell and 0–50 Hz for the accelerometer) were used to combine the load cell and wrist accelerometer data. This result demonstrates the advantage of integrating both sensor modalities, enhancing the ability of the model to detect fall events, reduce misclassifications, and improve overall performance.

Fig. 4 shows confusion matrices detailing classification results for models using a load cell, a wrist accelerometer, and the multimodal approach (load cell + wrist accelerometer). Specifically, the confusion matrices show results from fold 4 of the five-fold cross-validation. For fold 4, the training dataset included 32 participants with 3072 samples, from which 33 samples were removed owing to quality concerns, leaving 3039 samples for training. The testing dataset had eight participants with 768 samples, from which four samples were excluded, leaving 764 samples for testing (381 Non-fall and 383 Fall samples). The load cell model (Fig. 4(a)) achieved 79.97% accuracy, correctly classifying 88.98% of normal events but only 71.02% of Falls, with 28.98% of Falls misclassified as normal. The wrist-mounted accelerometer model (Fig. 4(b)) performed better with 95.16% accuracy, correctly classifying

Table 2

Averaged five-fold cross-validation performance for various frequency bands of accelerometer signals (mean $\pm$ standard deviation for test data).

Band (Hz)	Accuracy (%)	Sensitivity (%)	Specificity (%)
0–10	94.15 $\pm$ 3.98	91.41 $\pm$ 08.61	96.83 $\pm$ 1.84
0–20	94.07 $\pm$ 2.55	96.68 $\pm$ 01.29	92.24 $\pm$ 5.25
0–30	93.88 $\pm$ 3.93	92.66 $\pm$ 05.87	94.97 $\pm$ 3.07
0–40	93.99 $\pm$ 0.91	95.95 $\pm$ 02.43	92.55 $\pm$ 3.39
0–50	95.05 $\pm$ 1.18	95.64 $\pm$ 02.46	94.58 $\pm$ 1.66
10–20	89.56 $\pm$ 5.76	84.73 $\pm$ 13.10	94.19 $\pm$ 3.50
10–30	91.08 $\pm$ 4.78	87.88 $\pm$ 09.07	93.99 $\pm$ 3.35
10–40	85.71 $\pm$ 5.09	73.97 $\pm$ 11.81	97.06 $\pm$ 2.83
10–50	91.83 $\pm$ 4.17	87.65 $\pm$ 06.68	95.91 $\pm$ 4.75
20–30	85.53 $\pm$ 5.57	76.26 $\pm$ 12.98	94.28 $\pm$ 4.54
20–40	86.18 $\pm$ 4.40	77.51 $\pm$ 09.21	93.98 $\pm$ 3.27
20–50	85.84 $\pm$ 3.26	79.29 $\pm$ 06.47	91.39 $\pm$ 2.97
30–40	83.74 $\pm$ 4.65	74.59 $\pm$ 11.01	91.94 $\pm$ 4.74
30–50	84.06 $\pm$ 4.31	79.61 $\pm$ 11.94	87.81 $\pm$ 2.81
40–50	81.91 $\pm$ 7.32	74.46 $\pm$ 12.84	87.08 $\pm$ 4.70

Table 3

Comparison of model performance (mean $\pm$ standard deviation for test data) for each modality using its optimal frequency band: Load cell (0–10 Hz), Accelerometer (0–50 Hz), and Multimodal fusion (0–10 Hz load cell + 0–50 Hz accelerometer).

Signal	Accuracy (%)	Sensitivity (%)	Specificity (%)
Load cell (0–10 Hz)	74.44 $\pm$ 3.21	61.61 $\pm$ 4.89	82.99 $\pm$ 3.85
Accelerometer (wrist) (0–50 Hz)	95.05 $\pm$ 1.18	95.64 $\pm$ 2.46	94.58 $\pm$ 1.66
Load cell + Accelerometer (wrist)	96.26 $\pm$ 1.64	97.27 $\pm$ 1.48	95.21 $\pm$ 2.81

92.91% of normal events and 97.39% of falls. The multimodal approach (Fig. 4(c)) yielded the highest performance at 97.12% accuracy, correctly identifying 96.33% of normal events and 97.91% of falls. This improvement demonstrates the advantage of combining sensor modalities, enhancing the ability of the system to distinguish between Fall and Non-fall events, thus reducing misclassification while improving detection accuracy and sensitivity.

4.4. Training and validation evaluation

To assess the effectiveness and convergence behavior of our 1D CNN models, we visualized the training and validation loss and accuracy curves over 100 epochs for the load cell-only, accelerometer-only, and multimodal models (Fig. 5).

These curves represent the average performance across the five cross-validation folds, with shaded areas indicating standard deviation. The multimodal model showed the fastest convergence, closely followed by the accelerometer-only model, both consistently demonstrating low loss and high accuracy. The load cell-only model exhibited slower convergence, higher loss, and lower accuracy with greater variability.

Irrespective of the modality, the validation loss was minimized between epochs 40 and 60, after which it plateaued, whereas the training loss continued to decrease. A similar trend was observed in validation accuracy, which stabilized around the same epoch interval. In terms of accuracy, the accelerometer-only and multimodal models rapidly surpassed 90% within 20 epochs, with the multimodal model achieving the highest validation accuracy (95%). The load cell-only model reached approximately 70% accuracy around epoch 30. These results demonstrate that combining low-frequency load cell data with high-frequency accelerometer signals substantially enhances classification performance, balancing convergence speed and accuracy.

4.5. Signal analysis

4.5.1. Comparison of fall and non-fall conditions

Fig. 6 illustrates the PSD analysis of load cell signals across four sensors, comparing Fall and Non-fall conditions for all participants.

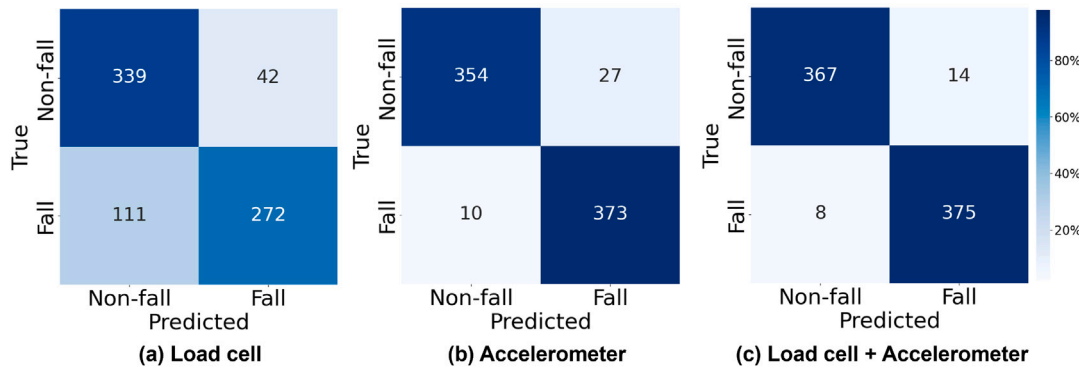


Fig. 4. Confusion matrix for the test dataset of sensor data in fourth fold of the five-fold cross-validation.

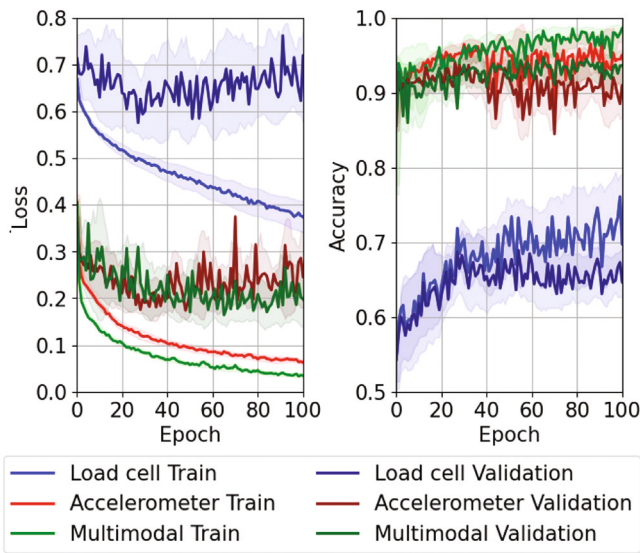


Fig. 5. Average training and validation loss (left) and accuracy (right) curves over 100 epochs for load cell-only, accelerometer-only, and multimodal (load cell + accelerometer) 1D CNN models, across five-fold cross-validation. Shaded areas indicate standard deviation across folds.

Blue lines and shaded intervals represent Fall signals' PSD, while red lines and intervals correspond to Non-fall signals. The shaded area indicates PSD standard deviation. Yellow highlights frequency ranges with significant differences ($p < 0.05$, Bonferroni-corrected). Across all load cells, significant PSD differences between Fall and Non-fall conditions were primarily observed in the 0–2 Hz range, emphasizing the importance of low-frequency components in distinguishing falls from Non-falls. Load cells 1 and 4, located at the right corner relative to bed exit, showed the most pronounced differences. The PSD characteristics discrepancies between conditions highlight that fall transitions generate stronger low-frequency components than Non-fall movements, such as normal bed exits. These differences may be critical for fall detection models using load cell data to identify Fall events.

Fig. 7 illustrates representative load cell signals for a example participant (S01), including (a) a simulated Fall event, (b) a Non-fall event, and (c) the time-domain difference between Fall and Non-fall signals. The shaded area (approximately 2.7–7.5-s) represents intervals where statistically significant differences (paired t-test, $p < 0.05$) between Fall and Non-fall events are consistently identified across all the load cells. Initially, both Fall and Non-fall signals showed minor fluctuations as the participant prepared to exit the bed. However, noticeable divergence in load distribution patterns emerged after the transition started (around 2-s). Specifically, load cells 2 and 3 (located

on the opposite side of the bed exit) exhibited greater signal magnitudes during Fall events, whereas load cells 1 and 4 (on the bed-exit side) showed lower magnitudes compared to those observed for the Non-fall events. These observations suggest that a more rapid and pronounced shift in weight distribution occurs during Fall scenarios, highlighting the potential of load cell signals to sensitively detect subtle yet distinct dynamic differences between fall and non-fall transitions.

4.5.2. Comparison of PSD across performance groups

Fig. 8 illustrates the accuracy distribution across participants, visualized as bar graphs, divided into three groups: Group 1 (high-performing, blue bars, 16 participants), Group 2 (intermediate-performing, orange bars, nine participants), and Group 3 (low-performing, green bars, 15 participants). The graph shows clear distinctions in accuracy levels between groups, with the performance of Group 1 remaining consistently higher than that of Group 3.

The spectral characteristics of these groups are shown in Fig. 9, in which the solid and white-dotted lines represent the mean PSDs of the Fall and Non-fall conditions, respectively; the blue and green lines correspond to Group 1 (high-performing participants) and Group 3 (low-performing participants), respectively. For both Fall and Non-fall signals, Group 1 exhibited higher PSD values across all load cells than Group 3, particularly in the 0–2 Hz range. This indicates that spectral energy in low-frequency bands is a critical feature of high-performance models. For Non-fall signals, Group 1 exhibited slightly higher PSD values in the 0–2 Hz range, although differences were less pronounced compared to fall signals.

Fig. 9(a) compares the PSDs of signals across groups (solid blue vs. solid green (Fall); dotted blue vs. dotted green (Non-fall)). Significantly different regions are shaded red for fall signals and blue for non-fall signals between Groups 1 and 3 ($p < 0.05$, Bonferroni-corrected). Across all load cells, significant differences in the 0–2 Hz range were observed between Group 1 and Group 3 signals for both Fall and Non-fall conditions. Load cells 2 and 4 showed broader spectral differences in the 0–10 Hz range for fall signals, suggesting that the high-performing group (Group 1) exhibited stronger spectral power in both narrow and broader low-frequency bands than the low-performing group (Group 3).

Fig. 9(b) compares the PSD of Fall and Non-fall signals within each group (same solid vs. dotted color). Shaded areas (red for Group 1 and blue for Group 3) indicate significant differences in frequency regions ($p < 0.05$, Bonferroni-corrected). For Group 1, significant differences were observed in the 0–2 Hz range across all load cells, emphasizing the importance of low-frequency components in distinguishing Fall and Non-fall signals for high-performance models. For Group 3, significant differences in the 0–2 Hz range were detected only for load cell 1, with minimal differences across other load cells. This highlights the reduced spectral separation between Fall and Non-fall signals for low-performing participants, particularly outside the 0–2 Hz range.

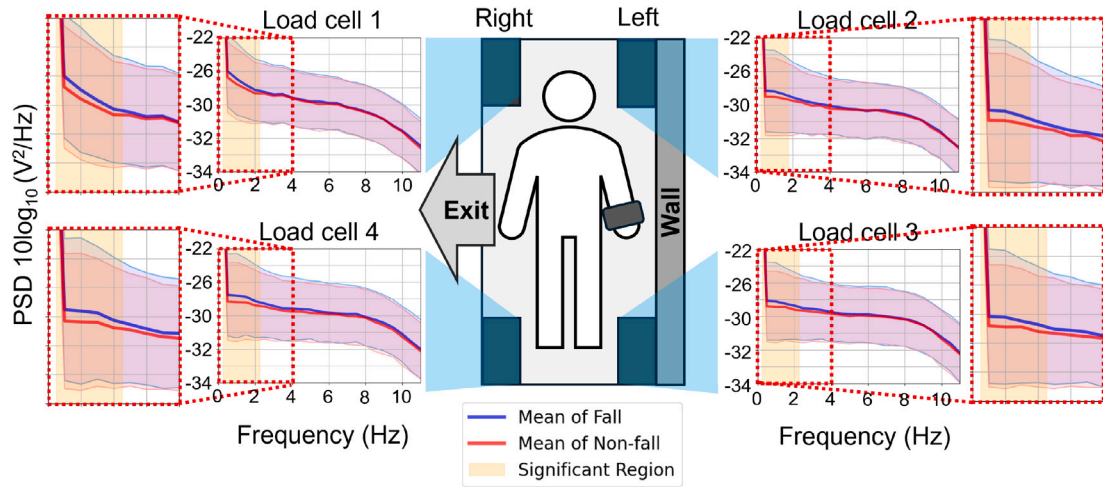


Fig. 6. PSD analysis comparing Fall and Non-fall signals across four load cells. The blue lines represent the mean PSD for Fall signals, and the red lines represent Non-fall signals. The shaded regions denote standard deviations, and the yellow-highlighted areas indicate significant differences between the conditions ($p < 0.05$, Bonferroni-corrected).

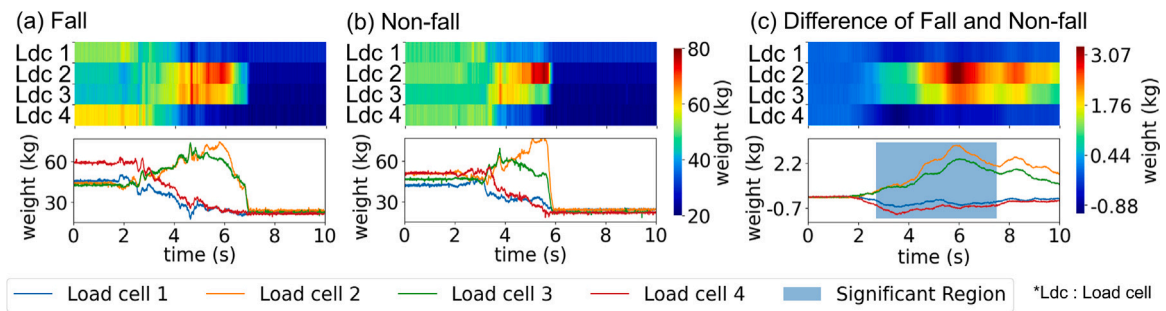


Fig. 7. Heatmap visualization of the four load cell signals. (a) Heatmaps and line plots of Fall data; (b) Heatmaps and line plots of Non-fall data; (c) Heatmaps and line plots illustrating differences between Fall and Non-fall conditions. Blue-shaded regions indicate time intervals with statistically significant differences between the two conditions ($p < 0.05$).

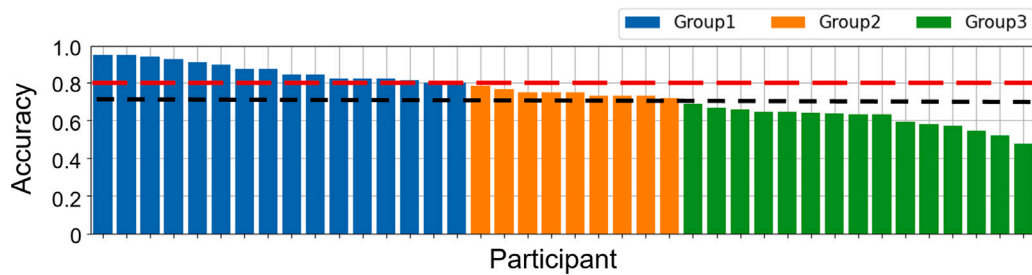


Fig. 8. Participant-wise fall detection accuracy grouped into three performance categories. The red dotted line represents the threshold between Groups 1 and 2 at 79.44%, and the black dotted line indicates the threshold between Groups 2 and 3 at 69.44%. Thresholds are determined based on the participants' average accuracy (74.44%), with a $\pm 5\%$ margin to ensure balanced group sizes and robust performance comparisons.

4.5.3. Correlation between performance and load cell spectral features

The correlation analysis results presented in Table 4 reveal the relationship between model accuracy and PSD differences in Fall and Non-fall signals across frequency bands for the four load cells. Significant positive correlations were found in the 0.5–2 Hz bands, with Load cell 4 showing the highest correlations (up to $r = 0.51$, $p < 0.05$), indicating its strong contribution to model accuracy. Load cells 2 and 3 also had significant correlations in this range, highlighting the importance of low-frequency spectral features. Higher frequency bands (>10 Hz) exhibited weak or negligible correlations, indicating limited utility for fall detection. Participant characteristics showed no significant correlations with model accuracy, such as height ($r = 0.04$, $p = 0.8026$), weight ($r = -0.14$, $p = 0.3721$), or BMI ($r = -0.22$, $p = 0.1757$).

These findings emphasize that participant demographics are not critical to model performance, reinforcing the importance of low-frequency signal properties in fall detection accuracy.

5. Discussion

5.1. Multimodal sensing system

In this study, we explored the feasibility of combining contactless bed-mounted sensors with wearable devices for fall detection. Bed-installed sensors provided real-time detection and user convenience, but showed limitations in detection accuracy, especially for complex fall scenarios. Wearable devices, particularly those with accelerometers

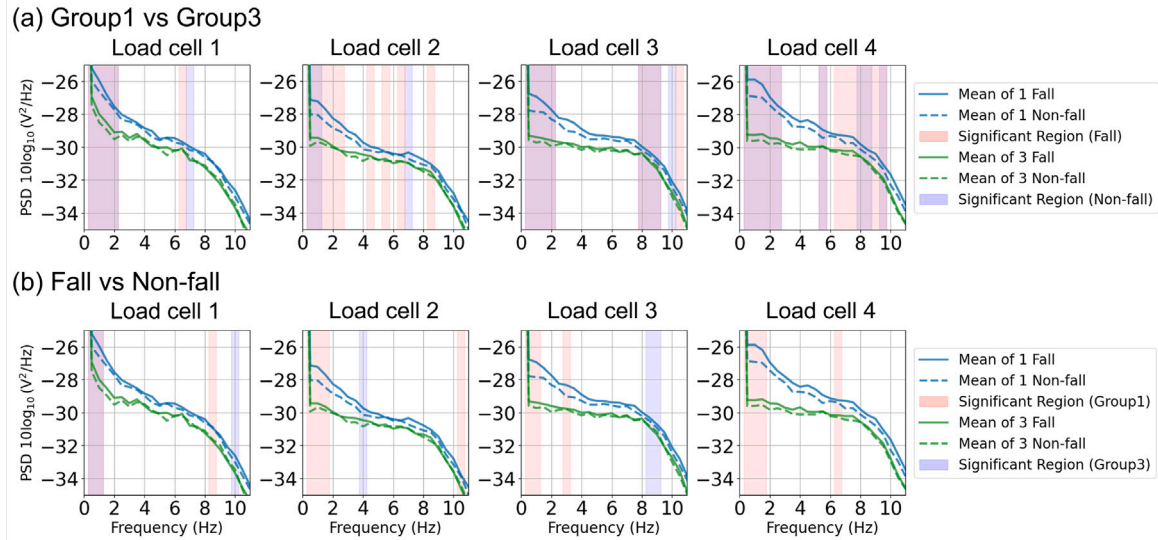


Fig. 9. PSD comparisons across performance groups. (a) PSD of Fall signals (solid lines) and Non-fall signals (dotted lines) compared between Group 1 (high-performing, blue) and Group 3 (low-performing, green). Shaded regions indicate frequency ranges with significant differences ($p < 0.05$, Bonferroni-corrected): red for Fall signals and blue for Non-fall signals. (b) PSD comparisons of Fall (solid lines) and Non-fall (dotted lines) signals within each group. The red and blue shaded areas highlight significant differences for Group 1 and 3, respectively.

Table 4

Correlation coefficients (r-values) between model accuracy and PSD differences (Fall vs. Non-fall) across load cells in various frequency bands.

Band (Hz)	Load cell1	Load cell2	Load cell3	Load cell4
0–0.5	0.14	−0.06	−0.04	0.02
0.5–1	0.36*	0.29	0.36*	0.35*
1–1.5	0.24	0.39*	0.32*	0.44*
1.5–2	0.23	0.38*	0.36*	0.51*
2–2.5	0.16	0.29	0.24	0.50*
0–2	0.24	0.05	0.10	0.14
0–10	0.26	0.07	0.13	0.17
0–20	0.31	0.10	0.09	0.13
0–30	0.30	0.12	0.07	0.12
0–40	0.25	0.10	0.05	0.10
0–50	0.24	0.05	0.08	0.12
10–20	0.40*	0.24	−0.18	−0.28
10–30	0.21	0.28	−0.20	−0.21
10–40	0.02	0.16	−0.21	−0.24
10–50	−0.07	−0.07	−0.12	−0.17
20–30	−0.07	0.17	−0.14	−0.04
20–40	−0.23	0.04	−0.16	−0.10
20–50	−0.26	−0.17	−0.03	−0.01
30–40	−0.26	−0.12	−0.13	−0.13
30–50	−0.26	0.03	0.06	0.02
40–50	−0.13	−0.06	0.20	0.02

* Indicates statistically significant correlations ($p < 0.05$).

like wrist-mounted smartwatches, demonstrate higher accuracy in fall detection. They are more effective in capturing detailed motion patterns, crucial for identifying falls. However, wearable devices are often removed for charging during sleep, interrupting continuous monitoring. Some users may find wearing a device overnight uncomfortable, potentially affecting adherence [37].

To address these challenges, we propose a hybrid framework integrating wearable devices and bed-mounted sensors for a flexible, adaptive system. This multimodal approach allows the strengths of both systems to complement each other [23]. Specifically, load cells and accelerometers provide complementary sensor information. Load cells at bed corners are highly sensitive to low-frequency signals (0–2 Hz), capturing subtle pressure distribution changes from body-weight shifts and postural adjustments [38]. Conversely, wrist-mounted accelerometers measure higher-frequency signals (up to 20 Hz), capturing rapid

limb movements and transient accelerations during fall events [39]. Furthermore, load cells infer patient movements through static load-distribution patterns, tracking body shifts on the bed, whereas accelerometers record dynamic wrist accelerations, providing kinematic details during motion events [40].

When users wear a smartwatch during sleep, the combination of wearable and bed sensor data enhances fall detection accuracy, providing robust coverage and early warnings of potential risks. Here, bed sensors act as a supplementary system to improve detection performance. Additionally, this integration reduces false alarms by cross-validating slow, low-frequency weight shifts from load cells with rapid acceleration changes from accelerometers [38,39]. Thus, the hybrid system more reliably distinguishes true fall events from benign movements like normal repositioning or stretching in bed.

Load cells provide essential advantages over wearable accelerometer-based systems. They require no user compliance, eliminating issues with device charging, battery life, and discomfort during sleep [37]. Load cells offer enhanced privacy compared to vision-based methods, recording only weight distribution without visual imagery. From a signal processing perspective, load cells capture low-frequency (0–2 Hz) energy peaks from gradual body-weight shifts, a critical feature distinguishing falls from non-fall events. Our frequency-domain analysis showed load cell signals at around 1 Hz had spectral differences (approximately 0.5–1 dB higher PSD in fall events), especially in bed-exit-side corners, supporting fall classification performance. Recent studies have demonstrated pressure-based sensors' effectiveness in capturing low-frequency signatures vital for accurate fall detection [38].

Although the accelerometer-only model achieved higher accuracy through direct motion measurement, incorporating load cell data improved overall accuracy by 2%, reducing false alarms. Moreover, load cells offer economic benefits, including simpler installation, minimal maintenance, and superior durability compared to pressure mats or vision-based systems, making them suitable for large-scale implementation in healthcare and residential facilities.

For users facing comfort issues with overnight wearable devices, bed-mounted sensors can provide a reasonable level of fall detection, although with slightly reduced accuracy. This hybrid system ensures continuous monitoring and protection, regardless of device-wearing preference. The goal is to develop a completely contactless system that facilitates accurate fall detection while maximizing user comfort

Table 5
Comparison of pressure sensor-based fall detection.

Author	Year	Type of sensors	Fall type	Number of record	Sampling rate (Hz)	Algorithm	Detection area W × H (cm)	Performance		
								Acc (%)	Sen (%)	Spe (%)
Zhang et al. [33]	2011	Multimodal (PM, PIR)	Floor	1	10	TH	79 × 54	–	88.89	100
Chaccour et al. [18]	2015	Air pressure	Floor	3	21 000	TH	100 × 100	–	88.80	94.90
Feng et al. [19]	2016	Fiber-optic	Floor	8	–	TH	35 × 35	97.67	–	–
Minvielle et al. [20]	2017	PE	Floor	28	100	RF	60 × 60	98.40	–	–
Yeo et al. [21]	2018	FSR	Floor	3	–	TH	100 × 200	90	–	–
Kowshik et al. ^a [34]	2020	Multimodal (PE, ACM)	Floor	–	–	–	–	–	–	–
Youngkong et al. ^b [35]	2021	PSU	Bedside	6	–	RF	90 × 190, 45 × 190	100	–	–
Wafiq et al. [36]	2023	FSR	Bedside	15	–	TH	40 × 80	92	–	–
Zhao et al. ^c [22]	2023	WDPDM	Bedside	30	100	TH	191 × 212	–	–	–
Proposed	2024	Load cell	Bedside	40	100	1D CNN	98 × 218	74.44	61.61	82.99
		Wearable (ACM)			1000		–	95.05	95.64	94.58
		Multimodal (load cell, ACM)			–		–	96.26	97.27	95.21

W: width, H: height, PM: pressure mat, PIR: passive infrared, PE: piezoelectric, FSR: force sensitive resistor, ACM: accelerometer, PSU: pressure-sensitive units, WDPDM: whole-torso distributed pressure detection mattress, TH: threshold, RF: random forest, Acc: accuracy, Sen: sensitivity, Spe: specificity.

^a This study proposes a framework, but fall detection experiments are not reported.

^b This study used two pressure mattresses, one on the bed and the other on the floor nearby bed.

^c This study proposed a fall detection algorithm that utilized pre-alarming time, false negatives, and false positives as performance metrics, achieving a pre-alarming time of 12.3, false negatives of 0, and false positives of 12.

and convenience. However, given the current limitations of contactless approaches realized using load cells, a hybrid system may serve as an effective intermediate solution. This approach broadens user options by providing both convenience- and performance-centered choices, paving the way for future advancements in contactless fall detection systems and their standalone applications.

5.2. Effective frequency bands for load cells

The optimal frequency range for detecting falls with load cells has not yet been well established. Accordingly, in this study, we aimed to identify the most effective frequency bands for fall detection using load cell signals because. Most studies on pressure sensors for fall detection focused on array-type sensors [18,20,33], typically using threshold-based approaches to detect falls based on triggered pressure points. However, these studies rarely explored the role of specific frequency bands in fall detection. Therefore, we applied various 5th-order Butterworth filters across different frequency bands, from 0–50 Hz in 10 Hz intervals, to investigate their effects on model performance. Informed by the Nyquist theorem, given the 100 Hz sampling rate of our load cell signals, all tested band ranges were below the 50 Hz limit. Our findings show that a frequency band of 0–10 Hz is critical for accurately distinguishing Fall events from normal bed exits using load cell data (Table 5).

Notably, the load cell sensors at the bed's four corners were sufficiently sensitive to capture subtle pressure changes during transitions, such as normal bed exits or simulated falls. Unlike array-type pressure sensors detecting falls based on pressure at specific grid points, our setup differentiated movements based on slight pressure distribution changes. Our results suggest the 0–10 Hz frequency band plays a crucial role in capturing dynamic aspects of bed exits and falls with load cells.

Interestingly, a difference was observed between optimal frequency bands for load cell (0–10 Hz, Table 1) and accelerometer data (0–50 Hz, Table 2). Although load cell performance was optimal in the 0–10 Hz range, statistical analysis (Fig. 6) revealed significant differences mainly within 0–2 Hz, highlighting the importance of ultra-low-frequency signals in pressure-based fall detection systems [38].

For accelerometer data, optimal performance was achieved in the 0–50 Hz band; however, removal of frequencies below 20 Hz caused a performance drop of approximately 85%, indicating the importance of the 0–20 Hz range for capturing human motion dynamics. This aligns with literature suggesting that IMU signals below 15–20 Hz represent the most significant components of human movements [39,40]. The

classification accuracy difference between 0–20 and 0–50 Hz bands was minimal (approximately 1%), suggesting that frequencies above 20 Hz contained marginally relevant information for fall detection tasks.

Overall, these optimal frequency sensitivities show the complementary roles of load cells and accelerometers in our multimodal fall detection approach. Load cells effectively capture slow pressure shifts critical to bed-exit events, while accelerometers capture rapid limb accelerations during falls. Combining these frequency sensitivities enhances fall detection accuracy. Future studies should investigate load cell signals within ultra-low-frequency ranges (0–2 Hz) and validate findings in real-world scenarios.

5.3. Signal analysis

The results highlight the spectral characteristics of load cell signals and their association with fall detection performance. A comprehensive examination of the existing literature indicates that several aspects warrant further in-depth investigations. Although the proposed hybrid system integrates load cell and accelerometer signals, this study is exclusively focused on load cell data for signal analysis. This decision has been motivated by the abundance of studies analyzing accelerometer signals for fall detection [41], whereas load cells remain underexplored. The primary goal is to validate the utility of load cells as a contactless sensor modality for fall detection and to distinguish their performance from accelerometer-based approaches. Future studies may involve analysis of accelerometer signals to identify complementary features that may enhance the performance of a hybrid system through multimodal data fusion.

While frequency-domain analysis identified significant differences between Fall and Non-fall conditions, temporal and spectro-temporal analyses have not yet been reported owing to variability in transition phases during the 30-s trials. The timing of transitions varied across trials, making it challenging to segment the signal into consistent temporal components. This limitation underscores the need for advanced techniques, such as explainable artificial intelligence methods and saliency maps or masking, to isolate key temporal features [42, 43]. Identifying these critical phases could provide deeper insight into fall dynamics, potentially improving classification performance and enabling predictive algorithms.

Correlation analysis revealed that low-frequency PSD differences (0.5–2 Hz) between Fall and Non-fall signals were positively associated with model accuracy. However, the observed correlation coefficients ($r \approx 0.3$ – 0.5) suggest that although these spectral features contribute to

performance improvement, they are not the sole determinants of model accuracy. This aligns with prior studies, indicating that other features, such as temporal or spatial signal patterns, may play an equally critical role in classification performance [44,45]. This suggests the need for further investigation of additional factors influencing model accuracy.

The group comparisons in Fig. 9 suggest directions for developing predictive fall detection biomarkers and enabling personalized monitoring strategies. High performers (Group 1) show significant differences in low-frequency bands (0–2 Hz), suggesting that these bands contain critical features for distinguishing fall events (Fig. 9(a)). Such characteristics can be isolated to explore biomarkers for fall risk assessment or enhance fall prediction capabilities of low performers. This analysis, highlighting the most useful frequency bands for classification, may aid in developing model optimization strategies, thus enabling improved signal preprocessing tailored to individual characteristics. Fig. 9(b) provides insights into group-level signal differences and guide patient-specific monitoring strategies. By analyzing PSD characteristics of bedside motion signals, clinicians can pre-screen patients to evaluate the predictive potential of load cell signals. If a patient's signals resemble those of the low-performing group (Group 3), then additional monitoring using accelerometers may be prioritized. Conversely, if signals align with those of the high-performing group (Group 1), then load cell data can be reliably utilized, potentially minimizing the need for wearable devices. This approach supports the development of personalized fall detection systems and tailors sensor's reliance via signal quality assessment.

5.4. Practical implications for real-world deployment

The hybrid fall detection system demonstrates substantial potential for practical deployment, providing effective monitoring particularly around patient beds, where approximately 60–70% of falls occur [8]. Load cells offer advantages for large-scale healthcare deployment, including durability, minimal maintenance, and cost-effectiveness compared to other systems such as full-scale pressure mats [46].

However, despite these strengths, the practical application of this system in real-world scenarios requires further investigation and development beyond its current form. In practical scenarios, patients and elderly individuals frequently move throughout their environment, not limited solely to bedside areas, thus necessitating monitoring solutions that can cover broader living and activity spaces.

In hospital environments, an effective system expansion could involve targeted installation of load cells beyond patient beds. Load cells could be embedded beneath caregiver beds, seating areas (chairs and benches), and critical transition zones like entrances to patient rooms and bathrooms. These strategic placements would broaden fall monitoring coverage and promptly alert medical personnel of falls, thus improving patient safety and emergency response times in clinical settings.

For home-care scenarios, our system could be extended by combining load cell-based monitoring with wearable accelerometers. During daytime activities, wearable accelerometers could monitor elderly individuals' movements, while load cell-based monitoring would provide continuous monitoring at night, particularly for those uncomfortable with overnight device usage [8,37]. Moreover, placing load cells beneath common household furniture — such as beds, sofas, armchairs, dining chairs, and tables — could economically extend fall detection coverage, capturing weight-shift patterns indicative of falls without requiring the high costs and extensive maintenance of comprehensive floor-based monitoring systems [47].

These envisioned practical deployment scenarios illustrate how the proposed hybrid system could realistically be extended to ensure comprehensive, reliable, and user-friendly fall detection across diverse environments. Future research should empirically validate these scenarios, focusing on sensor placements, integration methods, and system optimization to ensure maximum effectiveness and minimum false alarm rates in real-world applications.

5.5. Comparison of results with those of prior works

Table 5 presents a comparative analysis of the proposed fall detection system and existing pressure-sensor-based studies. The comparison includes sensor types, fall detection type (bedside or floor), classification algorithms, number of participants, sampling rates, detection area, and performance. Most studies employed a single sensor modality for fall detection, except Kowshik et al. [34] and Zhang et al. [33]. Kowshik et al. proposed a framework for multimodal fall detection using piezoelectric sensors and accelerometers, but did not include experimental results. Zhang et al. combined passive infrared sensors and piezoelectric microphones but only reported the integrated system's performance. In contrast, our study evaluates the individual performance of the load cell and accelerometer and demonstrates the performance improvement through a multimodal approach, offering greater flexibility for potential end-users.

A participant-number-focused analysis of existing literature indicates that only Minvielle et al. [20] and Zhao et al. [22] have involved more than 20 participants, whereas other studies, those of including Chaccour et al. [18] and Yeo et al. [21], have been conducted with the participation of less than 20 volunteers. Such small sample sizes raise concerns about the generalizability of these systems to real-world scenarios with diverse conditions. In our study, we addressed this limitation by recruiting 40 participants (20 males and 20 females), providing a sufficiently large sample size to improve generalizability compared to previous studies. This larger sample size aimed to reduce potential biases due to sex differences and strengthen the system's robustness across diverse fall scenarios.

A key distinction in the reviewed studies was the type of pressure sensors employed and their sampling rates. Chaccour et al. [18], used high-frequency rates of 21 kHz, whereas Zhang et al. [33] operated their proposed system at lower rates of 10 Hz. Minvielle et al. [20], Zhao et al. [22]. By contrast, we used a 100 Hz rate, balancing detailed fall-dynamics capture and computational efficiency. Notably, a significant number of previous studies have not reported their sampling rates, a critical omission as signal processing techniques and fall detection algorithms are often sensitive to sensor data frequency content.

Only Minvielle et al. [20] and Youngkong et al. [35] extracted features from pressure signals to classify falls using machine-learning methods like RF. Other studies mainly used simpler threshold-based methods, which may be less adaptable to variations in fall patterns or user-specific differences. Our approach introduces a deep-learning-based model, specifically a 1D CNN, allowing end-to-end learning without manual feature extraction. To our knowledge, this is the first application of a deep-learning-based method for fall detection using pressure sensors, providing a novel contribution. The use of 1D CNNs, well suited for time-series data, offers a promising avenue for improving the robustness and accuracy of fall detection systems.

Youngkong et al. [35] and Wafiq et al. [36] used pressure-sensitive arrays in mattresses or carpet-like materials, typically limited to specific environments. These pressure mats are often custom-designed to fit bed size, increasing costs and reducing scalability in real-world settings. Our system uses load cells installed at the bed frame's four corners, offering a more scalable and cost-effective solution. This design minimizes maintenance requirements as load cells are durable and less susceptible to wear than pressure-sensitive mats. Given its scalability and ease of installation, our system can be implemented in larger environments like hospitals with numerous beds, providing an economical and low-maintenance fall detection solution.

In summary, prior studies have demonstrated reasonable accuracy using pressure-sensor-based fall detection but have limitations related to small sample sizes, narrow detection areas, and simplistic algorithms. Our study proposes a multimodal, deep learning-based fall detection system with a larger participant pool, robust preprocessing methods, and greater scalability for real-world applications.

5.6. Limitations and future outlook

The data recorded in this study were based on simulated fall scenarios, not real-life falls. Collecting actual fall data presents ethical, safety challenges, and requires considerable resources and time. Ren et al. [48] reports that only 7.1% of fall detection studies using wearable devices employ real fall data, with no reported cases for nonwearable systems. Although simulated falls provide a practical solution for system validation, they may not fully capture real-world fall dynamics. We replicated fall-like actions and precursor behaviors, such as transitioning from lying to sitting on the bed edge, generating signals from load cells that mimicked actual falls. While visually distinguishing these signals was challenging, the data collection system captured subtle differences for the model to classify. Future studies should aim to collect real-world fall data in controlled environments, or observational studies should be conducted to improve the system's robustness and reliability.

To perform controlled experiments, we used an electronic trigger system that synchronized simulated falls, ensuring participants' movements followed specific instructions. Nevertheless, variations in participants' reaction times causes differences in the transition time, challenging robust algorithm development. For consistency, we segmented data into 30-s windows based on these triggers. While this method allows validation of our system in a controlled environment, it does not fully reflect real-world scenarios where falls occur unexpectedly without prior indication. Future studies should focus on developing asynchronous detection methods using a sliding-window approach that dynamically analyzes real-time data without relying on predefined triggers. Refining this asynchronous method could create a more adaptable and accurate fall detection system suitable for real-world deployment.

The proposed fall detection system can be applied in a real-world setting if it can function as an on-device solution, with a detection algorithm embedded in the hardware. This functionality can be realized via real-time data collection, minimal preprocessing, and local task execution without extensive reliance on network infrastructure. Although advantageous for real-time fall detection, it poses significant challenges due to hardware limitations. Complex deep learning models require substantial computational resources often unavailable in resource-constrained embedded systems [49]. To address this limitation, model optimization techniques, such as model compression and quantization, that reduce computational overhead and ensure efficient operation of low-power devices should be applied [50,51]. Alternatively, high-performance embedded systems could be considered, although this may increase costs and render them less feasible for widespread adoption. Balancing performance, computational efficiency, and cost-effectiveness is crucial for successful on-device fall detection.

Additionally, although in this study, we evaluated fall detection across simulated bed-exit scenarios from six postures, real-life bed movements such as waking up, stretching, breathing, or repositioning — activities not indicative of falls — were not explicitly tested. Such activities might generate low-frequency signals similar to those during laying down, potentially leading to false alarms and highlighting a limitation for real-world applicability. Further analyses using adaptive thresholding and artifact rejection techniques are essential to minimize false positives and enhance system robustness. Future studies should evaluate load cell-based systems during various real-world activities and investigate optimal sensor placements beyond bed areas to ensure comprehensive fall detection coverage and reliable performance under realistic conditions.

Although the hybrid fall detection system demonstrated high performance, integrating it into real-world healthcare environments poses challenges. To maximize utility, the system must be integrated with Internet of Things infrastructure and hospital information systems for real-time monitoring and automated alerts. In elderly care settings,

falls often require immediate medical attention, and a fully integrated system can alert healthcare personnel or caregivers when a fall is detected [52]. This integration ensures prompt action to reduce the risk of complications from unattended falls. Future studies should focus on developing seamless communication between fall detection systems and healthcare networks for timely intervention.

This study is primarily focused on fall detection, and other aspects of contactless health monitoring, such as sleep stage classification and posture monitoring, remain underexplored. Future investigations should incorporate accelerometer and ballistocardiogram data into a comprehensive system addressing multiple health monitoring needs. Additionally, further improvements can be achieved via refining data segmentation or implementation of movement-detection algorithms that focus only on relevant motion data to enhance accuracy and computational efficiency.

Moreover, predictive algorithms that can predict falls based on behavioral patterns, potentially preventing incidents before they occur, should be explored. After device validation through controlled experiments, its performance in real-world setting should be evaluated through clinical trials. If the system underperforms, then the proposed algorithm should be refined for better real-world applicability. A fully functional on-device system can be developed by optimizing the CNN models and refining the hardware to detect and alert users of fall incidents in real time.

6. Conclusions

In this study, we developed and validated a hybrid fall detection system, integrating contactless load cells and wearable wrist accelerometers; the proposed system can address challenges in current technologies, including discomfort, privacy issues, and limitations in detecting falls during sleep or bed transitions. The hybrid approach was evaluated across sensor combinations and configurations, with 40 participants performing simulated Fall and Non-fall activities in six postures, providing a comprehensive dataset for model training and testing. We also investigated effective frequency bands for load cell signals, an unexplored area in fall detection research.

The results showed combining load cell and accelerometer data significantly improved detection accuracy (96.26%), precision (95.21%), and sensitivity (97.27%), highlighting the multimodal approach's efficacy. Evidently, optimizing load cell signal processing using various filter configurations provided crucial insights, with best performance observed in the 0–10 Hz band range. We identified the critical role of the 0–10 Hz frequency range, emphasizing the importance of tuning signal processing parameters for load cell-based fall detection systems.

Significant differences between Fall and Non-fall signals were observed in the 0–2 Hz range, with high-performing groups exhibiting stronger spectral energies in low-frequency bands. Correlation analysis revealed PSD features within 0.5–2 Hz range play a critical role in influencing model accuracy, emphasizing low-frequency signals' importance in enhancing fall detection performance. These findings underscore the necessity of optimizing low-frequency signal processing to improve fall detection systems' accuracy and reliability.

The contributions of this study include: (1) introducing and validating load cells for bedside fall detection, a novel approach in this domain; (2) developing a deep learning model (1D CNN) optimized for multimodal sensor data fusion, showing improved performance over single-sensor models; and (3) exploring the effective frequency to optimize load cell data detection performance.

Future studies should concentrate on optimizing the system for real-world deployment. This includes refining sensor placement, enhancing model efficiency for on-device processing, and integrating the system into healthcare Internet of Things frameworks for seamless monitoring and automatic alert generation in clinical environments. These improvements aim to provide a comprehensive solution for proactive fall management, patient safety, and timely intervention.

CRediT authorship contribution statement

Sunghan Lee: Visualization, Methodology, Data curation, Writing – review & editing, Validation, Investigation, Conceptualization, Writing – original draft, Software, Formal analysis. **Jinwon Kim:** Writing – review & editing, Validation, Investigation, Conceptualization, Writing – original draft, Software, Formal analysis, Visualization, Methodology, Data curation. **In cheol Jeong:** Supervision, Methodology, Writing – review & editing, Resources, Funding acquisition, Validation, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.measurement.2025.118759>.

Data availability

Data will be made available on request.

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